

Air Conditioners Data Analysis

Stage 4: Data Validation & Cleaning

This notebook is independently executable in Google Colab. Every step explains what we do, why we do it, what we obtain, and how the result is used in the next stage.

Stage Objective

Stage 4 validates the dataset and prepares a reliable analytical DataFrame.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the project CSV directly so this notebook is independently executable.
csv_path1 = "/content/Most_Expensive_Products.csv"
df1 = pd.read_csv(csv_path1)
csv_path2 = "/content/Top_Discounted_Products.csv"
df2 = pd.read_csv(csv_path2)
csv_path3 = "/content/Top_Rated_Products.csv"
df3 = pd.read_csv(csv_path3)

print("CSV loaded successfully.")
print("Dataset shape:", df1.shape)
print("Dataset shape:", df2.shape)
print("Dataset shape:", df3.shape)
```

```
CSV loaded successfully.
Dataset shape: (15, 14)
Dataset shape: (15, 14)
Dataset shape: (15, 14)
```

Why Validation Is Important

A result can be numerically correct but still misleading if missing, duplicated, malformed or inconsistent values are not checked.

```
print('Total missing:', int(df1.isna().sum().sum()))
print('Total missing:', int(df2.isna().sum().sum()))
print('Total missing:', int(df3.isna().sum().sum()))
```

```
Total missing: 8
Total missing: 13
Total missing: 4
```

Reload the CSV

The source CSV is loaded directly to keep the stage reproducible.

```
print('Duplicates:', int(df1.duplicated().sum()))
print('Duplicates:', int(df2.duplicated().sum()))
print('Duplicates:', int(df3.duplicated().sum()))
```

```
Duplicates: 0
Duplicates: 0
Duplicates: 0
```

Missing-Value Check

Every column is checked for missing values. The exact counts are reported rather than assumed.

```
nums=df1.select_dtypes(include=np.number).columns.tolist(); print(nums)
nums=df2.select_dtypes(include=np.number).columns.tolist(); print(nums)
nums=df3.select_dtypes(include=np.number).columns.tolist(); print(nums)
```

```
['ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', 'tonnage', 'energy_star']
['ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', 'tonnage', 'energy_star']
['ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', 'tonnage', 'energy_star']
```

Duplicate Check

The complete row is checked for duplicates so repeated records can be identified before aggregation.

```
for c in nums: print(c, 'min=', df1[c].min(), 'max=', df1[c].max())
for c in nums: print(c, 'min=', df2[c].min(), 'max=', df2[c].max())
for c in nums: print(c, 'min=', df3[c].min(), 'max=', df3[c].max())
```

```
ratings min= 1.0 max= 5.0
no_of_ratings min= 1.0 max= 18.0
discount_price min= 71990.0 max= 128800.0
actual_price min= 77900.0 max= 149000.0
discount_amount min= 2000.0 max= 29150.0
discount_percent min= 2.5673940949935816 max= 23.65953109072375
tonnage min= 2.0 max= 4.0
energy_star min= 2.0 max= 5.0
ratings min= 3.0 max= 4.6
no_of_ratings min= 1.0 max= 1666.0
discount_price min= 199.0 max= 38999.0
actual_price min= 499.0 max= 89999.0
discount_amount min= 300.0 max= 51000.0
discount_percent min= 50.00416701391782 max= 60.120240480961925
```

```

tonnage min= 0.8 max= 2.0
energy_star min= 3.0 max= 5.0
ratings min= 5.0 max= 5.0
no_of_ratings min= 1.0 max= 2.0
discount_price min= 599.0 max= 128800.0
actual_price min= 999.0 max= 149000.0
discount_amount min= 400.0 max= 29660.0
discount_percent min= 12.349883491665173 max= 41.25173852573018
tonnage min= 1.0 max= 4.0
energy_star min= 3.0 max= 5.0

```

✓ Numeric Conversion Check

Numeric fields are safely converted where necessary and conversion failures are inspected.

```

clean_df1=df1.copy()
for c in nums: clean_df1[c]=pd.to_numeric(clean_df1[c],errors='coerce')
print(clean_df1.dtypes)
clean_df2=df2.copy()
for c in nums: clean_df2[c]=pd.to_numeric(clean_df2[c],errors='coerce')
print(clean_df2.dtypes)
clean_df3=df3.copy()
for c in nums: clean_df3[c]=pd.to_numeric(clean_df3[c],errors='coerce')
print(clean_df3.dtypes)

```

```

name                object
main_category       object
sub_category        object
image              object
link               object
ratings            float64
no_of_ratings       float64
discount_price      float64
actual_price        float64
discount_amount     float64
discount_percent    float64
tonnage            float64
energy_star         float64
brand              object
dtype: object
name                object
main_category       object
sub_category        object
image              object
link               object
ratings            float64
no_of_ratings       float64
discount_price      float64
actual_price        float64
discount_amount     float64
discount_percent    float64
tonnage            float64
energy_star         float64
brand              object
dtype: object
name                object
main_category       object
sub_category        object
image              object
link               object
ratings            float64
no_of_ratings       float64
discount_price      float64
actual_price        float64
discount_amount     float64
discount_percent    float64
tonnage            float64
energy_star         float64
brand              object
dtype: object

```

✓ Range Inspection

Minimum and maximum values are checked for numeric fields. This helps identify unexpected ranges that deserve attention.

```

for c in clean_df1.select_dtypes(include=['object','category']).columns: clean_df1[c]=clean_df1[c].astype('string').str.strip()
print('Text standardization done.')
for c in clean_df2.select_dtypes(include=['object','category']).columns: clean_df2[c]=clean_df2[c].astype('string').str.strip()
print('Text standardization done.')
for c in clean_df3.select_dtypes(include=['object','category']).columns: clean_df3[c]=clean_df3[c].astype('string').str.strip()
print('Text standardization done.')

```

```

Text standardization done.
Text standardization done.
Text standardization done.

```

✓ Text Consistency

Text/category fields are stripped of unnecessary leading and trailing spaces where appropriate so equivalent labels are less likely to split into separate groups.

```

print(clean_df1.shape)
print(clean_df2.shape)
print(clean_df3.shape)

```

```

(15, 14)
(15, 14)
(15, 14)

```

✓ Clean Working DataFrame

A working DataFrame is created after the validation operations. The original CSV is not overwritten.

```
print('Remaining missing:', int(clean_df1.isna().sum().sum()))
print('Remaining missing:', int(clean_df2.isna().sum().sum()))
print('Remaining missing:', int(clean_df3.isna().sum().sum()))
```

```
Remaining missing: 8
Remaining missing: 13
Remaining missing: 4
```

✓ Quality Result

The final quality checks are displayed again so the reader can see the state of the data after validation.

```
print(clean_df1.head())
print(clean_df2.head())
print(clean_df3.head())
```

```
      name main_category \
0  Voltas 3Ph Scroll SAC C2-NVenture I-Cassette A...  appliances
1  Voltas Venture i Cassette 3Ph Scroll SAC(C2-N)...  appliances
2    Voltas Venture Slimline Tower AC (3 Ton White)  appliances
3  Blue Star 2.0 Ton 3 Star Inverter Cassette SAC...  appliances
4  0 General 2 Ton 5 Star Inverter Split AC (Copp...  appliances

      sub_category      image \
0  Air Conditioners https://m.media-amazon.com/images/I/31lyjr30Xs...
1  Air Conditioners https://m.media-amazon.com/images/W/IMAGERENDE...
2  Air Conditioners https://m.media-amazon.com/images/I/31jhp14GmP...
3  Air Conditioners https://m.media-amazon.com/images/I/41slFsUdX0...
4  Air Conditioners https://m.media-amazon.com/images/I/611TN01YAg...

      link  ratings  no_of_ratings \
0  https://www.amazon.in/Voltas-Scroll-SAC-C2-NVE...      5.0      2.0
1  https://www.amazon.in/Voltas-3-0T-Venturei-Cas...      4.0     18.0
2  https://www.amazon.in/Voltas-Venture-Slimline-...      3.2      5.0
3  https://www.amazon.in/Blue-Star-Inverter-Casse...      4.0     18.0
4  https://www.amazon.in/General-Inverter-Copper-...      5.0      1.0

      discount_price  actual_price  discount_amount  discount_percent  tonnage \
0      128800.0      149000.0      20200.0      13.557047      4.0
1      102800.0      120490.0      17690.0      14.681716      3.0
2      99750.0      128900.0      29150.0      22.614430      3.0
3      87800.0      109800.0      22000.0      20.036430      2.0
4      82260.0      85260.0      3000.0      3.518649      2.0

      energy_star  brand
0      NaN      Voltas
1      NaN      Voltas
2      NaN      Voltas
3      3.0  Blue Star
4      5.0    0 General

      name main_category \
0  RISARYA H20R1203 IGBT 1PCS POWER TRANSISTOR 20...  appliances
1      CNC Split AC Stand (2 Tons, Off-White)  appliances
2  LG 1.0 Ton 5 Star DUAL Inverter Wi-Fi Window A...  appliances
3  ALLWIN TAFTA High Quality Metal Special Coated...  appliances
4  iMee-1.5 Ton Inverter Split Air Conditioner wi...  appliances

      sub_category      image \
0  Air Conditioners https://m.media-amazon.com/images/W/IMAGERENDE...
1  Air Conditioners https://m.media-amazon.com/images/I/71tkqWth8...
2  Air Conditioners https://m.media-amazon.com/images/I/71WuTUFvml...
3  Air Conditioners https://m.media-amazon.com/images/I/61gx0sUfF3...
4  Air Conditioners https://m.media-amazon.com/images/W/IMAGERENDE...

      link  ratings  no_of_ratings \
0  https://www.amazon.in/RISARYA-H20R1203-TRANSIS...      4.0      4.0
1  https://www.amazon.in/CNC-Split-AC-Stand-Off-W...      4.0     18.0
2  https://www.amazon.in/LG-Inverter-Convertible-...      4.2    1208.0
3  https://www.amazon.in/ALLWIN-Quality-Mounted-S...      4.6      7.0
4  https://www.amazon.in/iMee-1-5-Conditioner-Pro...      4.0     18.0

      discount_price  actual_price  discount_amount  discount_percent  tonnage \
0      199.0      499.0      300.0      60.120240      NaN
1      800.0     2000.0     1200.0      60.000000      2.0
```